

Resturant Management System

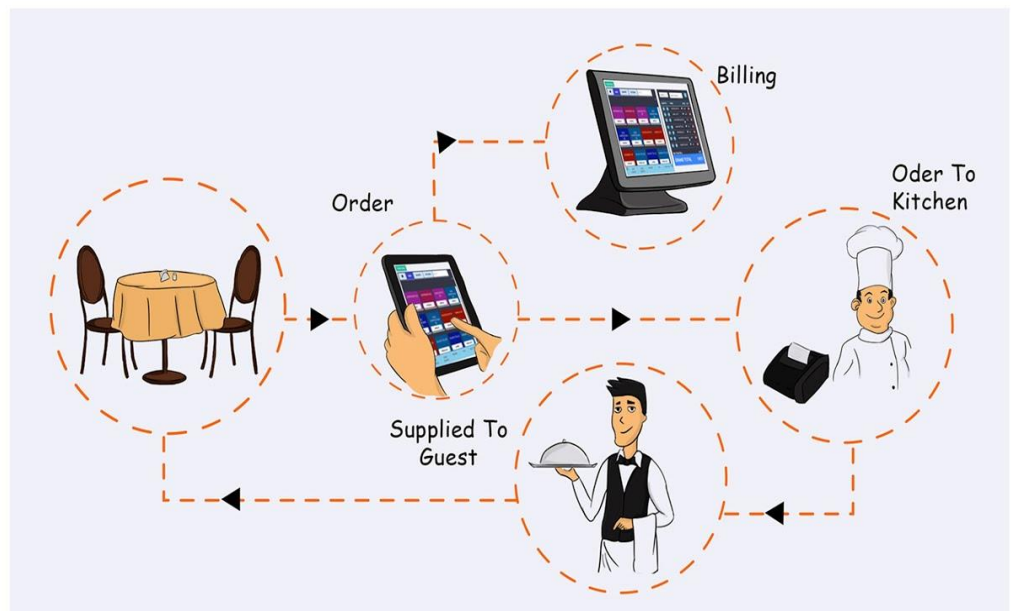


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Description :

The Restaurant Management System (RMS) facilitates seamless collaboration among various stakeholders within a restaurant, ensuring operational efficiency and customer satisfaction. Customers interact with the RMS to browse the menu, place orders, and track their order statuses in real-time, enhancing their dining experience. They can also make reservations for dining and provide feedback through the integrated review system. Restaurant Managers utilize the system to update restaurant details , employees details and check feedbacks to track performance metrics. Waitstaff use the system to record and manage customer orders, ensuring accurate and timely service. Chefs rely on the inventory module to monitor ingredient levels, update stock usage, and request replenishments. Cashiers handle payment processing via multiple methods such as cash, credit cards, and digital wallets, ensuring a smooth checkout process. Delivery Personnel schedule deliveries, update delivery statuses, and provide real-time tracking for takeout orders.

Features:

Restaurant Information Management:

- Display and update restaurant details like name, address, phone number, and email.

Menu Management:

- Display menu items categorized by type.

Add, update, or remove menu items.

- Manage menu categories with descriptions.

Order Management:

- Place and track orders with status updates.
- Manage order items with references to menu items.
- Calculate total costs of orders.

Customer Management:

- Display and update customer information.
- Maintain and review customer order history.
- Allow customers to place orders and make reservations.

Employee Management:

- Display and update employee details such as name, role, and shift schedules.
- Manage orders assigned to employees.

Payment Processing:

- Handle payments through various methods like cash, card, and digital wallets.
- Update and track payment statuses.

Inventory Management:

- Monitor and update inventory stock levels.
- Manage inventory items with quantities and unit prices.
- Generate inventory reports and manage stock replenishments.

Delivery Management:

- Schedule and update delivery details like address and time.
- Provide real-time tracking for deliveries.

Reservation System:

- Make and update reservations with dates and times.
- Associate reservations with customer details.

Report Generation:

- Generate detailed reports on sales, inventory, and performance.
- Display and export report data as needed.

Feedback System:

- Collect and manage customer feedback.
- Use feedback to enhance service quality.

Data Analytics:

- Analyze sales trends, popular items, and operational efficiency.
- Optimize operations using data insights.

Classes And Their Attributes:**1. RMS****Attributes:**

RMSID: int

Name: String

Location: String

Methods:

AddRMS(): void

DeleteRMS(): void

UpdateRMS(): void

GetRMS(): RMS

2. Person

Attributes:

PersonID: int

Name: String

Address: String

PhoneNo: String

Methods:

AddPerson(): void

DeletePerson(): void

UpdatePerson(): void

GetPerson(): Person

3. Customer

Attributes:

CustomerID: int

LoyaltyPoints: int

Methods:

AddCustomer(): void

DeleteCustomer(): void

UpdateCustomer(): void

GetCustomer(): Customer

EarnLoyaltyPoints(): void

RedeemLoyaltyPoints(): void

4. Employee

Attributes:

EmployeeID: int

JobTitle: String

Methods:

AddEmployee(): void

DeleteEmployee(): void

UpdateEmployee(): void

GetEmployee(): Employee

5. Admin

Attributes:

AdminID: int

Methods:

AddAdmin(): void

DeleteAdmin(): void

UpdateAdmin(): void

GetAdmin(): Admin

6. Menu

Attributes:

MenuID: int

Methods:

AddMenuItem(): void

DeleteMenuItem(): void

UpdateMenuItem(): void

GetMenuItem(): Menu

7. Order

Attributes:

OrderID: int

OrderDate: Date

TotalAmount: double

Methods:

CreateOrder(): void

CancelOrder(): void

UpdateOrder(): void

GetOrder(): Order

8. Item**Attributes:**

ItemID: int

ItemName: String

ItemPrice: double

Methods:

AddItem(): void

RemoveItem(): void

UpdateItem(): void

GetItemDetails(): Item

9. Payment**Attributes:**

PaymentID: int

Amount: double

PaymentMethod: String

Methods:

ProcessPayment(): void

GetPaymentDetails(): Payment

RefundPayment(): void

10. Cash**Attributes:**

CashAmount: double

Methods:

MakeCashPayment(): void

11. Card**Attributes:**

CardNumber: String

ExpiryDate: String

CVV: String

Methods:

MakeCardPayment(): void

Uml Diagram:

